

# Ananthi Manokaran

## FPGA RTL ENGINEER

Experienced FPGA RTL Engineer with almost four years of expertise in designing, implementing, and optimizing digital systems using VHDL and Verilog. Proficient in leveraging Xilinx FPGA development tools, adept at debugging and simulation, and skilled in collaborative teamwork to consistently deliver high-quality results within specified timelines. Committed to continuous learning and eager to contribute to cutting-edge FPGA development projects.



## Work History

### Senior Executive Engineer

**Pinaka Aerospace Solutions Pvt. Ltd., Bangalore**      **April'23 – Present**

**Client : Aeronautical Development Establishment, DRDO**

- Developed VHDL code to establish transmission and reception between Holt 8582 IC using ARINC 429 protocol.
- Implemented VHDL code for handling ARINC format data transmission and reception using control word register value from microcontroller.
- Conducted rigorous debugging of the program through simulation to ensure its functionality and reliability.

### FPGA Engineer

**Ciyes Systems Pvt. Ltd., Ciombatore**

**Oct'22 - Feb'23**

- Spearheaded the maintenance of a completed project, ensuring its continued functionality and efficiency.
- Provided comprehensive training to junior team members in digital design and Verilog.
- created comprehensive training material for junior team members in digital design and Verilog, including the documents necessary for their training.
- Demonstrated a proactive approach by acquiring foundational knowledge in Python to facilitate its use in upcoming projects.
- Conducted in-depth exploration and experimentation with the Kria K260 Vision Kit to support project development.
- Gained valuable insights into quantization techniques, which are crucial for custom AI model development.



## Contact

### Address

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### LinkedIn

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## Skill set

- Digital Design
- RTL Coding
- Simulation
- STA
- CDC
- Test bench creation
- Chipscope debugging
- ILA, ICON
- Oscilloscope



## Education

### • BE Electronics and Communication Engineering

- Sri Ramakrishna Engineering College, Coimbatore, 2015-19
- 7.34 CGPA

## Design Engineer

**BLRLABS Pvt.Ltd.**, Bengalore

**Dec'21 - Apr'22**

- Successfully managed the maintenance of previously completed projects and associated documentation.
- Gained valuable experience in implementing crucial aspects of physical design, including conducting sanity checks and performing floor planning to ensure optimal project performance and functionality.

## FPGA Engineer

**Appasamy Associates.**, Chennai

**Mar'19 - Dec'21**

- Designed a graphical user interface (GUI) by harnessing RTL design techniques to enhance user interaction and system control.
- Established seamless communication between hardware and software components through the UART protocol, promoting effective data exchange and system integration.
- Developed comprehensive test benches to validate and simulate the entire codebase, ensuring its reliability and performance.
- Implemented data storage in SPI Flash memory, enhancing data persistence and retrieval.
- Created custom fonts using Verilog, contributing to the project's unique and appealing visual elements.
- Streamlined data flow within SPI Flash memory using test benches, eliminating the need for additional hardware components.
- Successfully incorporated the ability to read multiple BMP images from an SD card and display them, enhancing visual capabilities.
- Utilized UART communication to dynamically change displayed images based on user-defined modes.
- Seamlessly inserted images into specified locations on the display, enhancing the overall user experience.

## Board used

- Spartan 6 9lx
- Alinx AX7020
- Zybo z7
- Zedboard
- Alinx Spartan 45lx

## Protocol known

- SPI
- I2C
- UART
- AMBA,AXI
- SD Card
- ARINC 429

### • 12<sup>th</sup> Stare Board

- Govt Girls Higher Secondary School, Thondamuthur, 2014-15
- 81.16 %

### • 10<sup>th</sup> CBSE

- Govt Girls Higher Secondary School, Thondamuthur, 2012-13
- 93 %



## Languages

- Tamil
- English
- Kannada



## Software

- Vivado 2022
- ISE 14.7
- Vitis / SDK



## Programming Languages

- VHDL
- Verilog
- C programming
- Pyhon



## Project

| Title                                                                                | Organizaion                                                                           | Period          |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------|
| <ul style="list-style-type: none"><li>Arinc 429 protocol</li></ul>                   | <ul style="list-style-type: none"><li>Pinaka aerospace solutions, Bangalore</li></ul> | Apr'23 – Sep'23 |
| <ul style="list-style-type: none"><li>PHACO GALAXY LEO – Image overlay</li></ul>     | <ul style="list-style-type: none"><li>Appasamy associates,Chennai</li></ul>           | jun'21 – Dec'21 |
| <ul style="list-style-type: none"><li>NON CONTACT TONOMETR – Font creation</li></ul> | <ul style="list-style-type: none"><li>Appasamy associates,Chennai</li></ul>           | May'20 – May'21 |
| <ul style="list-style-type: none"><li>PHACO GALAXY LEO – Video overlay</li></ul>     | <ul style="list-style-type: none"><li>Appasamy associates,Chennai</li></ul>           | May'19 – Apr'20 |



## Interests/Hobbies

- Gardening
- Crocheting
- Reading
- Singing
- Dancing